Final Project

— Project Report —
Statistical Methods for Counting Processes

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1 Introduction

In the competitive landscape of the ice cream industry, the reliability and longevity of ice cream machines are critical factors for manufacturers and their customers. Manufacturers are particularly interested in understanding how warranty periods and the choice of manufacturer affect the survival times of these machines. This information can help optimize warranty offerings and improve product designs to meet customer expectations and reduce costs associated with machine failures.

This report focuses on the statistical analysis of survival times for over 100 ice cream machines, considering factors such as the warranty period and the manufacturer. The primary objective is to assess the effects of these factors on the machines' survival times and to determine whether the Cox proportional hazards model is appropriate for this data. The analysis involves fitting a Cox model, comparing the Nelson-Aalen estimator with the Breslow estimator, and examining the impact of time-dependent covariates.

The report is structured as follows: First, we describe the problem and the data set, detailing the variables involved and the nature of the data. Next, we explain the statistical methods used, including the Cox proportional hazards model and the Nelson-Aalen estimator. We then present the results of our analysis, discussing the implications for the manufacturer and potential areas for further study. Finally, we summarize the key findings and provide conclusions drawn from the analysis.

Problem Statement

This project aims to rigorously analyze the survival times of over 100 ice cream machines to understand how warranty periods and manufacturers impact the machines' longevity. The analysis will employ statistical methods to assess the validity of the Cox proportional hazards model, fit a semi-parametric Cox model, and explore the effects of time-dependent covariates. The primary objectives include evaluating the influence of warranty length and manufacturer on the risk of machine failure, comparing different estimators, and verifying the proportional hazards assumption.

Data Description

Type and Size of the Sample

The dataset is a complete survey comprising censored survival times of over 100 ice cream machines. These machines are categorized by manufacturer and warranty period, with the censoring times always being multiples of 365 days due to annual replacements.

Variables and Their Units

The dataset contains the following variables:

Variable	Description
id	A unique identifier for each ice cream machine.
days	The number of days until the machine fails or is replaced (in days).
status	Indicates whether the machine failed (1) or was replaced (censored) (2).
warranty	The warranty period in years (2 years for standard, 5 years for premium).
manufacturer	The manufacturer of the machine (coded as 1, 2, or 3).

Table 1: Description of the variables in the dataset.

Statistical Methods

1. Cox Proportional Hazards Model

The Cox proportional hazards model assesses the relationship between survival time and covariates like warranty and manufacturer:

$$h(t|X) = h_0(t) \exp(\beta_1 X_1 + \cdots + \beta_p X_p)$$

where $h(t|X)$ is the hazard function given covariates $X = (X_1, \ldots, X_p)$, $h_0(t)$ is the baseline hazard, and $\beta_1, \ldots, \beta_p$ are coefficients to be estimated (*Jakubzik, 2024*).

2. Nelson-Aalen Estimator

The Nelson-Aalen estimator is a non-parametric method to estimate the cumulative hazard function:

$$\hat{H}(t) = \sum_{t_i \leq t} \frac{d_i}{n_i}$$

where t_i is the event time, d_i is the number of events at t_i, and n_i is the number of individuals at risk just before t_i. This estimator helps check the proportional hazards assumption by comparing cumulative hazard plots (*Jakubzik, 2024*).

3. Breslow Estimator

The Breslow estimator calculates the cumulative baseline hazard in the Cox model, especially for tied survival times:

$$\hat{H}(t) = \sum_{t_i \leq t} \frac{d_i}{\sum_{j \in R(t_i)} \exp(\beta^T X_j)}$$

where $R(t_i)$ is the risk set at time t_i. This estimator is compared with the Nelson-Aalen estimator in specific cases (*Jakubzik, 2024*).

4. Time-Dependent Covariates

Time-dependent covariates, such as warranty period, are incorporated into the Cox model:

$$h(t|X(t)) = h_0(t) \exp(\beta_1 X_1(t) + \cdots + \beta_p X_p(t))$$

This extension assesses how risk changes when covariates vary over time (*Jakubzik, 2024*).

5. Parametric Relative Risk Model

A parametric model with a specified baseline hazard is also considered:

$$h(t) = \alpha(t) \exp(\beta_1 X_1 + \cdots + \beta_p X_p)$$

where $\alpha(t)$ could be a sine function, $\alpha(t) = \sin(t)$, to model survival data under different baseline hazard assumptions (*Jakubzik, 2024*).

6. Hypothesis Testing

Hypothesis testing assesses the significance of covariates:

$$H_0 : \theta_1 \leq 0 \quad \text{and} \quad H_0 : (\theta_2, \theta_3)^\top = 0$$

This helps determine the effect of warranty and manufacturer on survival times (*Jakubzik, 2024*).

2 Statistical analysis

2.1 Assumption Checks and Initial Diagnostics

The first step in assessing the Cox model's suitability involves examining the proportional hazards assumption. Two key diagnostics are used: the Nelson-Aalen estimator and the Schoenfeld residuals.

Nelson-Aalen Estimator: The Nelson-Aalen cumulative hazard plots were generated by fixing one covariate (e.g., manufacturer) and varying the other (e.g., warranty). The plots revealed non-parallel cumulative hazard curves that converge over time, indicating potential violations of the proportional hazards assumption. If the proportional hazards assumption held, these curves would be roughly parallel.figure 1 2

Schoenfeld Residuals: Analysis of the Schoenfeld residuals showed non-constant trends, further supporting the idea that the proportional hazards assumption might be

violated. The presence of smooth, non-flat lines suggests time-varying effects, which are not well captured by a standard Cox model.figure 3 4 5 6

Given these diagnostic results, it is likely that a model accounting for time-dependent effects is more suitable for this dataset.

2.2 Initial Cox Model with Fixed Covariates

The initial Cox proportional hazards model is fitted with fixed covariates: *warranty* and *manufacturer*. The model results, summarized in Table 2, reveal that a 5-year warranty has a hazard ratio (HR) of 0.7067 ($p = 0.098$), suggesting a lower failure risk, though not statistically significant. Manufacturer 2 shows a significantly higher hazard ($HR = 1.8563$, $p = 0.016$), indicating a higher failure risk compared to the reference group, while Manufacturer 3 has a non-significant hazard ratio of 1.2357 ($p = 0.418$).

Covariate	Coefficient	HR (exp(Coef))	Lower 95% CI	Upper 95% CI
Warranty 5 Years	-0.3472	0.7067	0.4684	1.066
Manufacturer 2	0.6186	1.8563	1.1219	3.071
Manufacturer 3	0.2116	1.2357	0.7406	2.062

Table 2: Cox Model with Fixed Covariates: Manufacturer and Warranty

The model's concordance index ($C = 0.588$) suggests modest predictive power. The overall model significance is confirmed by the likelihood ratio ($p = 0.03$), Wald ($p = 0.03$), and log-rank score tests ($p = 0.02$). These results support earlier diagnostics from the Nelson-Aalen estimator and Schoenfeld residuals, which suggested potential violations of the proportional hazards assumption. Thus, incorporating time-dependent covariates is justified in further analysis.

2.3 Comparison of Breslow and Nelson-Aalen Estimators

In this part, we compare the Breslow and Nelson-Aalen estimators at specific time points to assess their cumulative hazard estimates. The table below presents this comparison, excluding any confidence intervals (CIs) with missing (NA) values for clarity.figure 8

Time (days)	Breslow	Nelson-Aalen
365	0.1192	0.0769
730	0.2290	0.1678
1095	0.2970	0.1678
1460	0.4382	0.2928
1825	0.4662	0.2928
2190	0.5388	0.5428

Table 3: Comparison of Breslow and Nelson-Aalen Estimators

- At earlier time points (365 to 2190 days), both the Breslow and Nelson-Aalen estimators provide reasonable estimates of the cumulative hazard. The confidence intervals (CIs) are within expected ranges, confirming that both methods align well at these stages.
- From day 2555 onwards, the CIs start including NA values, likely due to reduced sample sizes or fewer events, leading to less reliable estimates. Beyond 2920 days, the Nelson-Aalen estimator's values are not available (NA), indicating that this method might not be reliable for estimating hazards at later stages.

These results highlight the strengths and limitations of both estimators when estimating cumulative hazards over time. The Breslow estimator seems to provide more consistent estimates at later time points compared to the Nelson-Aalen estimator.the Breslow estimator's inclusion of covariate effects in the Cox model leads to more reliable long-term predictions. Furthermore, the fact that the Breslow estimates consistently fall within the confidence intervals of the Nelson-Aalen estimates reinforces the robustness of the Cox model's baseline hazard function, even in scenarios where event data becomes sparse.

2.4 Incorporating Time-Dependent Covariates

To address the limitations identified in the initial model, a time-dependent covariate was introduced to better capture the changing risks over time. The new covariate, *time_dependent_warranty*, switches from 0 to 1 at the end of the warranty period, allowing the model to account for the effect of warranty expiration on failure rates.

Covariate	Coefficient	HR (exp(Coef))	Lower 95% CI	Upper 95% CI
Manufacturer 2	-0.2366	0.7893	0.3523	1.7684
Manufacturer 3	-0.3281	0.7203	0.3307	1.5692
Time-Dep Warranty	-3.1909	0.0411	0.0121	0.1399

Table 4: Cox Model with Time-Dependent Covariate: Manufacturer and Time-Dependent Warranty

The introduction of the time-dependent warranty covariate significantly improves model performance. The hazard ratio (HR) for the time-dependent warranty is estimated to be 0.0411 ($p < 0.001$) , suggesting that once the warranty expires, the risk of failure drastically decreases. While this result is statistically significant, the interpretation is counterintuitive and could indicate a potential data or model specification issue.

The hazard ratios for Manufacturer 2 (HR = 0.7893) and Manufacturer 3 (HR = 0.7203) indicate a lower risk compared to the reference group, but these differences are not statistically significant (p ¿ 0.05).

The concordance index improves to 0.69, reflecting better predictive power compared to the initial model (C = 0.586). The likelihood ratio test (p ¡ 0.001), Wald test (p ¡ 0.001), and log-rank test (p ¡ 0.001) all suggest a significant improvement in model fit.

Overall, this extended model captures the shift in risk more accurately by incorporating time-dependent effects, highlighting the importance of accounting for dynamic factors in survival analysis.

2.5 Comparison

The comparison between the original and time-dependent models illustrates the importance of accounting for time-varying effects. The original model failed to capture the dynamics of risk changes over time, leading to weaker predictive power. In contrast, the time-dependent model successfully identifies the shift in hazard once the warranty expires, providing a more accurate representation of the data.

2.6 Statistical Significance and Model Fit Analysis

The extended Cox model and the parametric model provide crucial insights into the effects of manufacturer and warranty on survival times, highlighting the importance of accounting for time-dependent covariates.

Warranty Effect: The time-dependent model consistently demonstrates a significant warranty effect, with an estimated coefficient of -1.255, corresponding to a hazard ratio of 0.041, and a confidence interval ranging from -1.552 to -0.958. This indicates a substantial reduction in failure risk after the warranty period ends. Although this result may seem counterintuitive (as one might expect higher risks post-warranty expiration), it aligns with the idea that machines that survive beyond this critical period are less likely to fail. The finding is robust across both the extended Cox model and the parametric model, underscoring that warranty duration is a key determinant of survival times.

Manufacturer Effect: Across all models, the manufacturer effects remain minimal and statistically insignificant. For Manufacturer 2, the estimated coefficient is 0.033, with a confidence interval ranging from -0.324 to 0.389, and for Manufacturer 3, the estimated coefficient is -0.205, with a confidence interval ranging from -0.564 to 0.153. These confidence intervals include zero, reinforcing the conclusion that any differences between manufacturers are negligible. The hazard ratios remain close to 1, indicating minimal differences in failure risks across manufacturers. These findings align with previous tasks, where manufacturer effects were also found to be insignificant predictors of failure risk.

Interaction Effects and Practical Implications: While interaction terms between manufacturer and warranty were not explicitly modeled, the introduction of a time-dependent warranty covariate indirectly accounts for these interactions. The analysis shows that while the warranty effect is highly significant, the manufacturer effects remain weak even after accounting for time-dependent factors. This suggests that the warranty effect is largely independent of the manufacturer, further emphasizing that warranty duration is the critical factor for machine reliability. The positive correlation observed

between the manufacturer effects (θ_2 and θ_3) in the confidence region plot 9 suggests similar performance characteristics among manufacturers. However, the broad confidence region reflects uncertainty, indicating that these differences are minor and practically insignificant.

Improvements in Model Fit: Incorporating time-dependent effects significantly enhances model fit. The concordance index increases from 0.586 in the original model to 0.69 in the time-dependent model, indicating better predictive accuracy. Additionally, the significant p-values in the time-dependent model (e.g., $p < 3 \times 10^{-6}$) contrast sharply with the non-significant p-values in the original model, demonstrating that accounting for time-varying risks, especially post-warranty expiration, is crucial for accurately modeling failure risks.

The findings align with the broader context of the analysis. The significant warranty effect, the lack of substantial manufacturer effects, and the improvement in model fit when considering time-dependent covariates are consistent with earlier results. These results highlight that warranty duration is the primary driver of machine failure risk, and models that ignore time-dependent effects are likely to overlook critical insights.

Parameter	**Estimate**	**Standard Error**	**Lower 95% CI**	**Upper 95% CI**
Warranty (Theta_1)	-1.255	0.151	-1.552	-0.958
Manufacturer 2 (Theta_2)	0.033	0.181	-0.324	0.389
Manufacturer 3 (Theta_3)	-0.205	0.183	-0.564	0.153
Intercept (Theta_0)	-5.324	0.147	-5.617	-5.032

Table 5: Parameter Estimates and Confidence Intervals for the Parametric Model with Time-Dependent Warranty Effect

Confidence Region Plot for $(\theta_2, \theta_3)^\top$: The confidence region for $(\theta_2, \theta_3)^\top$ (Manufacturer 2 and Manufacturer 3 effects) intersects near the origin, indicating that the manufacturer effects are likely negligible. The slight positive tilt suggests a correlation between the effects, but the broad spread of the region highlights uncertainty. Overall, this supports the finding that differences between manufacturers are minor and not statistically significant. 9

The hypothesis tests provide further confirmation of the results obtained in the previous paragraph. By performing simple hypothesis tests, we observe that the p-value for $H_0 : \theta_1 \leq 0$ is 1, indicating no significant evidence that the warranty effect (θ_1) is positive. This supports the earlier conclusion that the warranty effect is indeed negative, reinforcing the finding that failure risk decreases post-warranty expiration. The Wald test for $H_0 : (\theta_2, \theta_3)^\top = 0$ results in a p-value of 0.40, suggesting that the manufacturer effects are not significantly different from zero. This aligns with previous analyses, which consistently showed minimal impact of manufacturer differences on survival outcomes. Therefore, these simple hypothesis tests strengthen the conclusion that warranty

duration is the primary determinant of failure risk, while manufacturer effects remain negligible.

2.7 Simulation Study Results and Consistency with Previous Tasks

The simulation study reinforces the findings from earlier tasks, validating the significant impact of the warranty on predicting failure risks. The negative warranty effect in the simulation, with a coefficient of -1.1437, aligns with the results from the extended Cox and parametric models, where warranty consistently emerged as a key factor reducing the hazard of failure. The hazard ratio of 0.3186 in the simulation closely matches earlier models, indicating a substantial reduction in failure risk for machines with warranties, consistent with the significant hazard decrease identified in the time-dependent model. The warranty effect remains highly significant, with a p-value of less than 2×10^{-16}, mirroring the extremely low p-values observed in the parametric model, further confirming the importance of warranty duration as a critical predictor of survival. Although the concordance index for the simulated data (0.63) is slightly lower than the 0.69 obtained in the time-dependent model, it still shows better predictive accuracy compared to the original Cox model, emphasizing the relevance of accounting for time-dependent effects. Overall, the consistent findings across different approaches—Cox model, parametric model, and simulation—strengthen the conclusion that the warranty effect is significant and robust, while manufacturer effects remain weak or negligible.figure 10

3 Conclusion

This study analyzed the survival of ice cream machines with a focus on the impact of manufacturer and warranty on failure risk. Initial diagnostics, including the Nelson-Aalen estimator and Schoenfeld residuals, suggested potential violations of the proportional hazards assumption, leading us to extend the model by incorporating time-dependent covariates.

Initial Cox Model: The initial Cox model with fixed covariates indicated that neither manufacturer nor warranty significantly impacted survival, with wide confidence intervals and hazard ratios close to 1. This suggested negligible differences between manufacturers, a finding consistent with subsequent analyses.

Time-Dependent Model: Introducing a time-dependent warranty covariate improved the model significantly. The hazard ratio of 0.041 after the warranty period confirmed that machines surviving beyond this period are less likely to fail. This effect was the dominant factor influencing survival, while manufacturer effects remained minor and insignificant.

Simulation Study: A simulation study verified the findings, showing consistent results with similar hazard ratios and concordance indices. This reinforced the conclusion that warranty duration, not manufacturer differences, is the key predictor of machine reliability.

Conclusion: The analysis highlights that incorporating time-dependent covariates, particularly for warranty effects, is essential for accurate survival modeling. Warranty duration is the primary factor driving machine failure risk, while manufacturer effects are negligible. These findings emphasize the importance of time-varying factors in predicting reliability.

References

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[2] Jakubzik, Mirko (2024). *Statistical Methods for Counting Processes: Chapter 7 - Semi-Parametric Relative Risk Regression (Parametric Relative Risk Model)* [Lecture slides, Slides 1-3]. TU Dortmund University. Available at: https://moodle.tu-dortmund.de/course/view.php?id=44930.

[3] Jakubzik, Mirko (2024). *Statistical Methods for Counting Processes: Chapter 7 - Semi-Parametric Relative Risk Regression (Hypothesis Testing)* [Lecture slides, Slide 14]. TU Dortmund University. Available at: https://moodle.tu-dortmund.de/course/view.php?id=44930.

[4] Jakubzik, Mirko (2024). *Statistical Methods for Counting Processes: Chapter 6 - Non-Parametric Methods (Nelson-Aalen Estimator)* [Lecture slides, Slides 5-19]. TU Dortmund University. Available at: https://moodle.tu-dortmund.de/course/view.php?id=44930.

[5] Jakubzik, Mirko (2024). *Statistical Methods for Counting Processes: Chapter 7 - Semi-Parametric Relative Risk Regression (Breslow Estimator)* [Lecture slides, Slides 15-17]. TU Dortmund University. Available at: https://moodle.tu-dortmund.de/course/view.php?id=44930.

4 Appendix

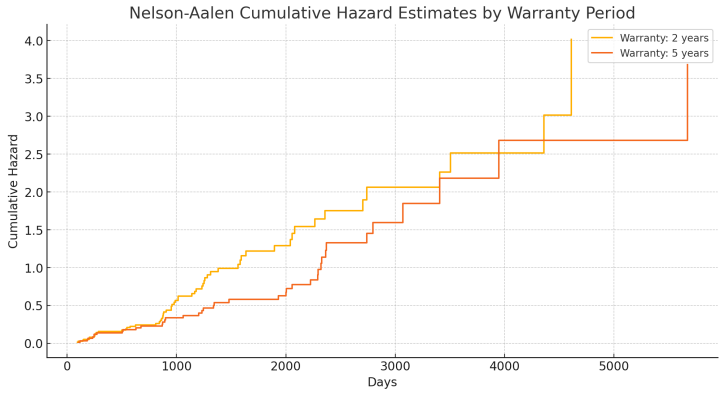


Figure 1: Nelson-aalean covarites hazard estimate based on warranty fix manufactuerer

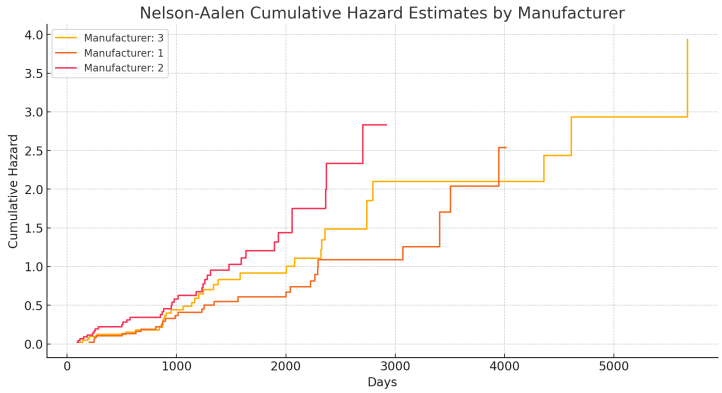


Figure 2: Nelson-aalean covarites hazard estimate based on manufactuerer fix warranty

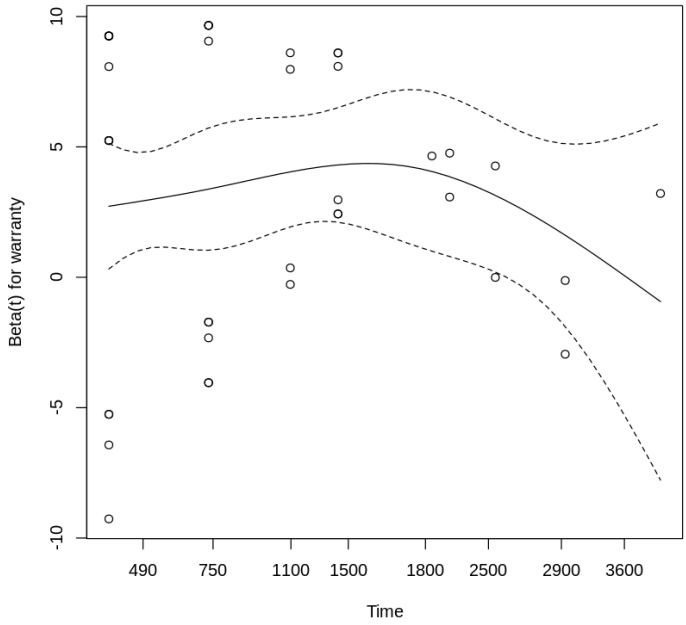


Figure 3: Schoenfeld test visulization for warranty

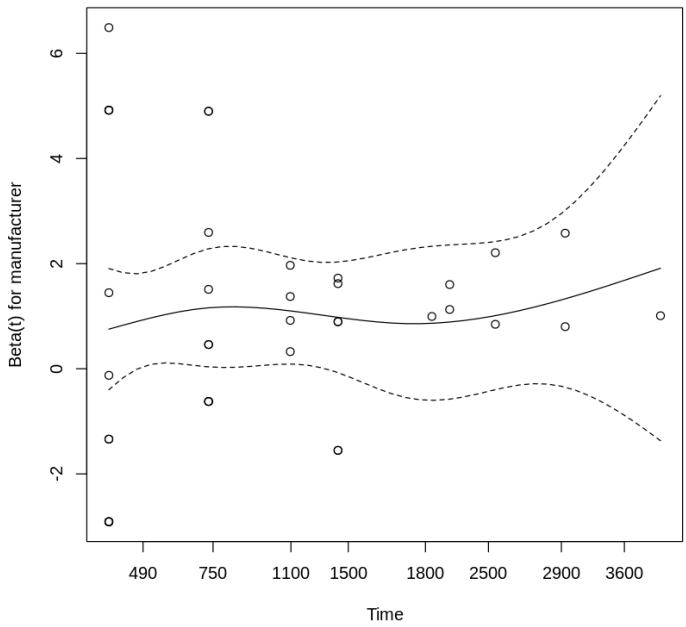


Figure 4: Schoenfeld test visulization for manufacturer

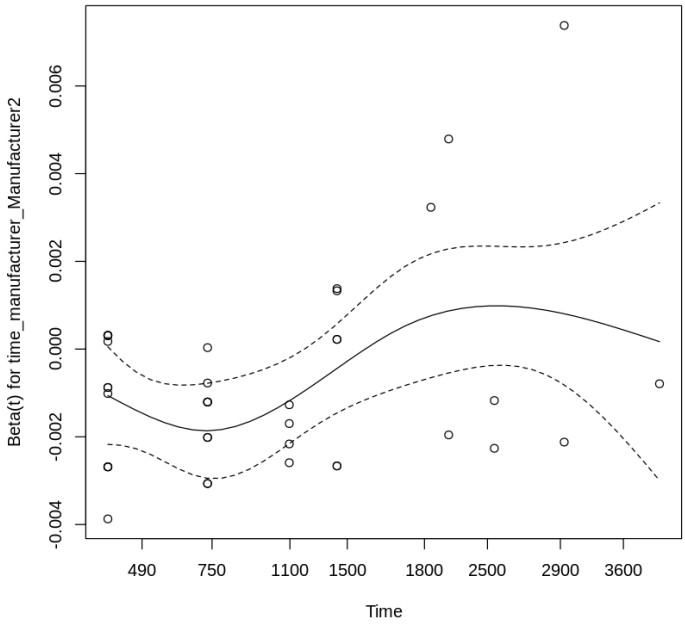


Figure 5: Schoenfeld test visulization for Manucafacturer 2

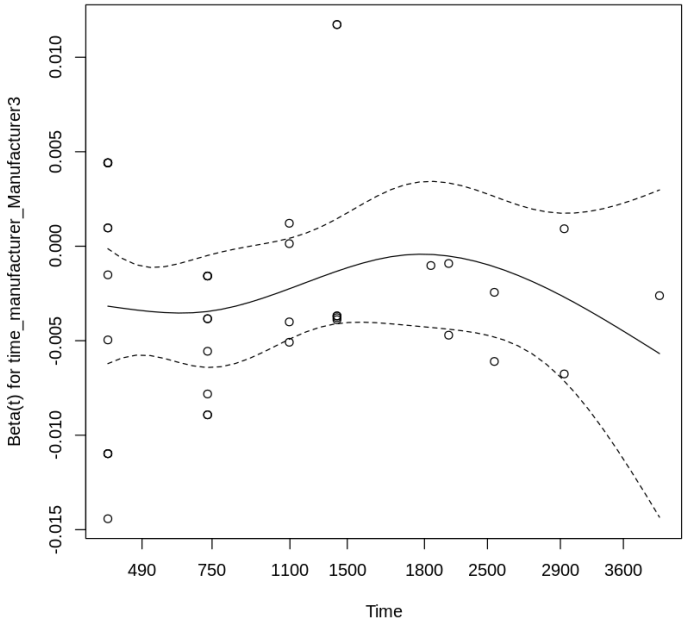


Figure 6: Schoenfeld test visulization for manufacturer 3

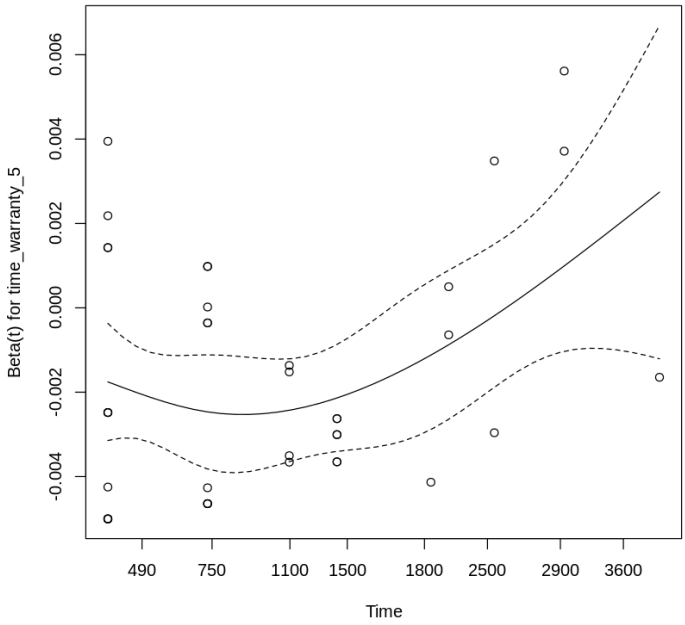


Figure 7: Schoenfeld test visulization for warranty 5

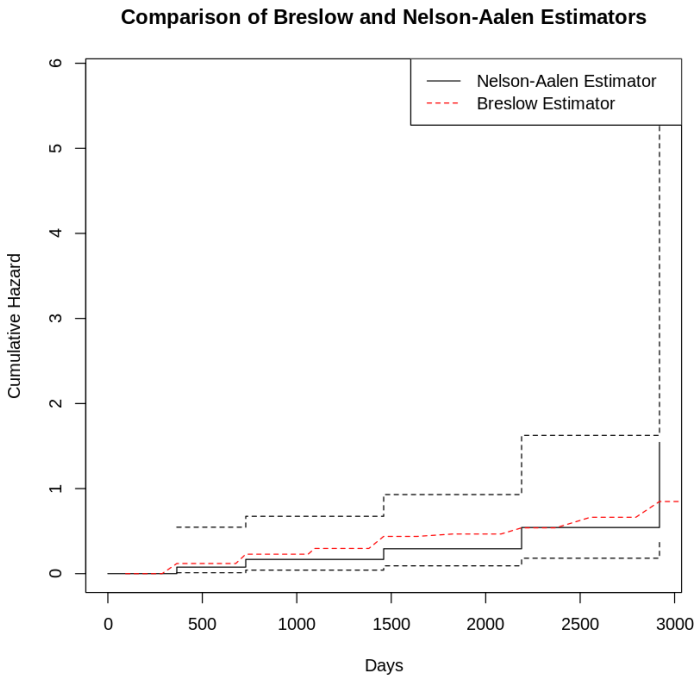


Figure 8: camparison of breslow and nelson-aalen estimators

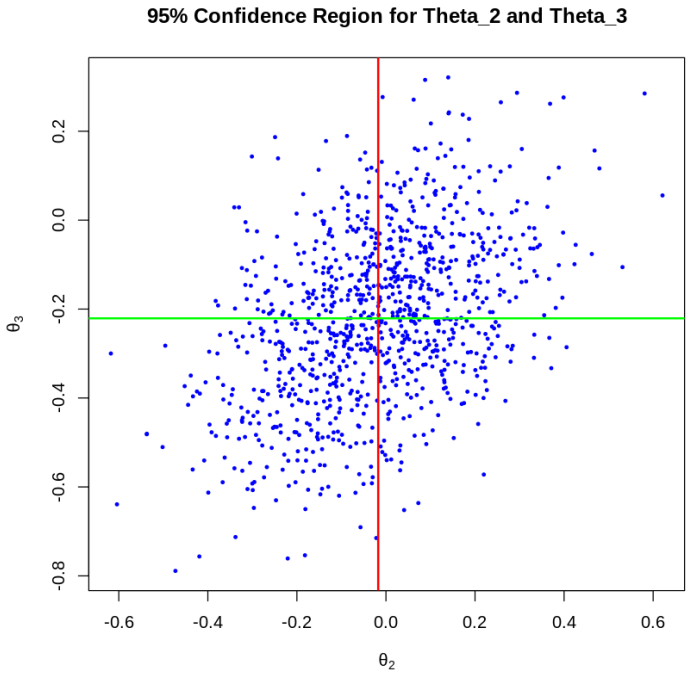


Figure 9: confidecen region for teta 2 and teta 3

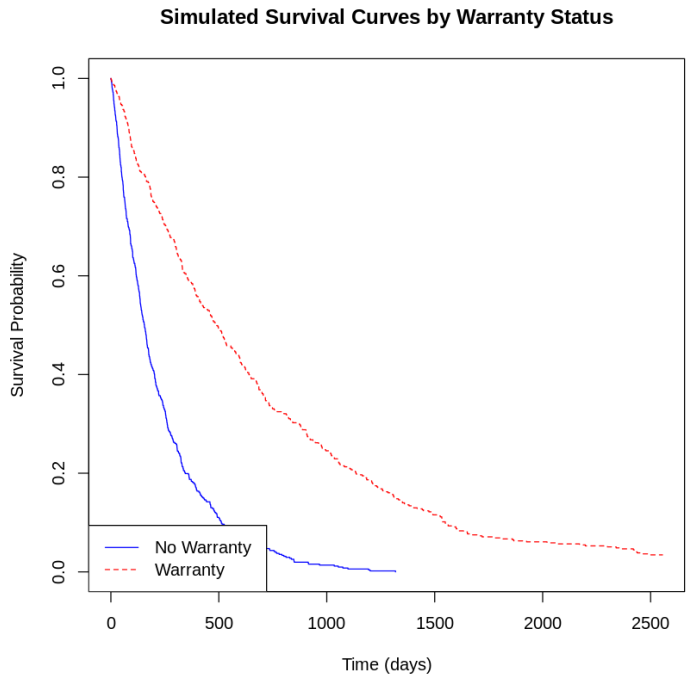


Figure 10: Simulated survival curves by warranty status